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Revolutionizing Site Investigation: Digitalizing Survey Data to Improve Seabed Characterization Accuracy and Optimize Project Cost Efficiency

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Abstract

An extensive geophysical and geotechnical database has been created from a large quantity of geophysical and geotechnical data acquired in ADNOC's offshore fields. The database was rigorously and carefully analyzed to develop a seabed zone map and PSI/CSI calculators to estimate monotonic Pipe-Seabed Interaction (PSI)/Cable-Seabed Interaction (CSI) parameters for preliminary pipeline and cable design. A data relative confidence level map was also developed to present the reliability of the defined seabed zones for the calculated PSI/CSI parameters. These maps and tools were integrated into ADNOC's ArcGIS platform to enable ADNOC's pipeline and cable engineers to make data-driven project decisions.

Keywords: Data digitization, Geographic Information System (GIS), Geophysical and geotechnical database, Pipe Seabed Interaction (PSI), Cable Seabed Interaction (CSI)

Introduction

Over the past few decades, Abu Dhabi National Oil Company (ADNOC) Offshore has acquired a large quantity of geophysical and geotechnical data in their offshore fields. In 2024, ADNOC Offshore initiated a Seabed Zonation Study to digitize and compile all the acquired geophysical and geotechnical data into a GIS database. [Fig. 1](#) presents the field area covered by the Seabed Zonation Study.

One of the main objectives of the study is to develop a seabed zone map based on the geophysical and geotechnical data acquired within the upper 1 meter of seabed across the study area to facilitate Pipeline/Cable-Seabed Interaction (PSI/CSI) assessment and to support pipeline and cable design. Additionally, the study also aims to develop tools for generating monotonic PSI/CSI parameters appropriate for preliminary design of rigid pipelines, flexible pipelines and cables to be installed within the study area.

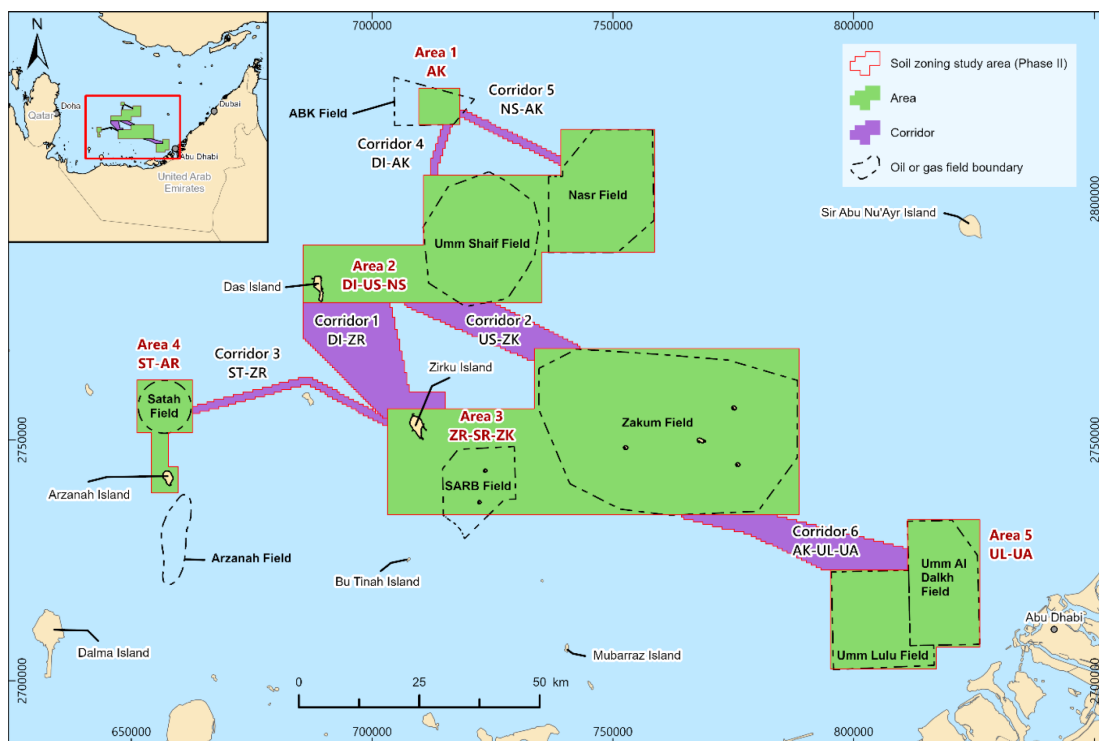


Figure 1—Study field area

The workflow adopted in the Seabed Zonation Study is shown on Fig. 2. This workflow was designed to enable rigorous and efficient analysis of the extensive geophysical and geotechnical dataset compiled in this study for (i) delineating seabed zones exhibiting similar geotechnical characteristics across the study area; and (ii) establishing a robust methodology for deriving monotonic PSI/CSI parameters that appropriately reflect seabed property variabilities and data uncertainties observed in the compiled dataset to support pipeline and cable design.



Figure 2—Seabed Zonation Study workflow

This paper presents the methodology used to:

- Digitize and integrate the geophysical and geotechnical dataset

- Score, screen and analyze the integrated geophysical and geotechnical dataset to develop a data relative confidence level map and a seabed zone map for PSI/CSI study purposes
- Select pipeline/cable inputs and geotechnical parameters for Monte Carlo PSI/CSI analyses
- Develop PSI/CSI calculators for generating monotonic PSI/CSI parameters appropriate for preliminary pipeline and cable design

Following the completion of this study, the compiled geophysical and geotechnical database, along with the seabed zone map, PSI/CSI calculators and Pipeline/Cable Alignment Segmentation Tool developed in this study were integrated into ADNOC's ArcGIS platform. This integration enables ADNOC's pipeline and cable engineers to improve future project planning and make data-driven decisions in their pipeline and cable design processes.

Data Compilation and Preparation

One of the main tasks of the Seabed Zonation Study was the compilation of geophysical and geotechnical data relevant for the study. A large quantity of geophysical and geotechnical data from earlier surveys, public sources, and consultant archives acquired in and near the study area were reviewed, compiled and integrated into a digital database. In addition to the compiled digital data, factual reports, drawings, and other pertinent documentation, a thorough review of ADNOC ArcGIS database was also undertaken to extract data relevant for this study. Sections below summarize the compiled geophysical and geotechnical database for the study.

Geophysical Data

Multi-beam echo sounder (MBES) bathymetry, side scan sonar (SSS), backscatter, and sub-bottom profiler (SBP) data were gathered from more than 60 surveys conducted in the study area between 1995 and 2023, covering approximately 94% of the 6,011 km² study area. Among the compiled geophysical data, bathymetric data has the highest coverage area of any available geophysical dataset and served as one of the fundamental data for interpreting seabed conditions in this study. The low resolution publicly available [GEBCO \(2023\)](#) bathymetry data that covers the entire study area (except land) was also used as a reference in regions where no bathymetry data was available. The coverage of the compiled geophysical data is presented in [Fig. 3](#) and summarized in [Table 1](#).

Table 1—Geophysical data coverage summary

Combination	Data coverage type	Coverage (%)
Areas considered for seabed characterisation		94.28
1	Bathymetry	27.99
2	Bathymetry & Side Scan Sonar	60.76
3	Bathymetry, Side Scan Sonar & Backscatter	1.24
Sub-total		89.99
4	Bathymetry, Sub-bottom Profiler & Side Scan Sonar	2.03
5	Bathymetry, Sub-bottom Profiler, Side Scan Sonar & Backscatter	2.26
Sub-total		4.29
Areas not considered for seabed characterisation		5.72
0	No geophysical data coverage	5.38
N.A.	Not applicable (islands)	0.34
Total		100.00

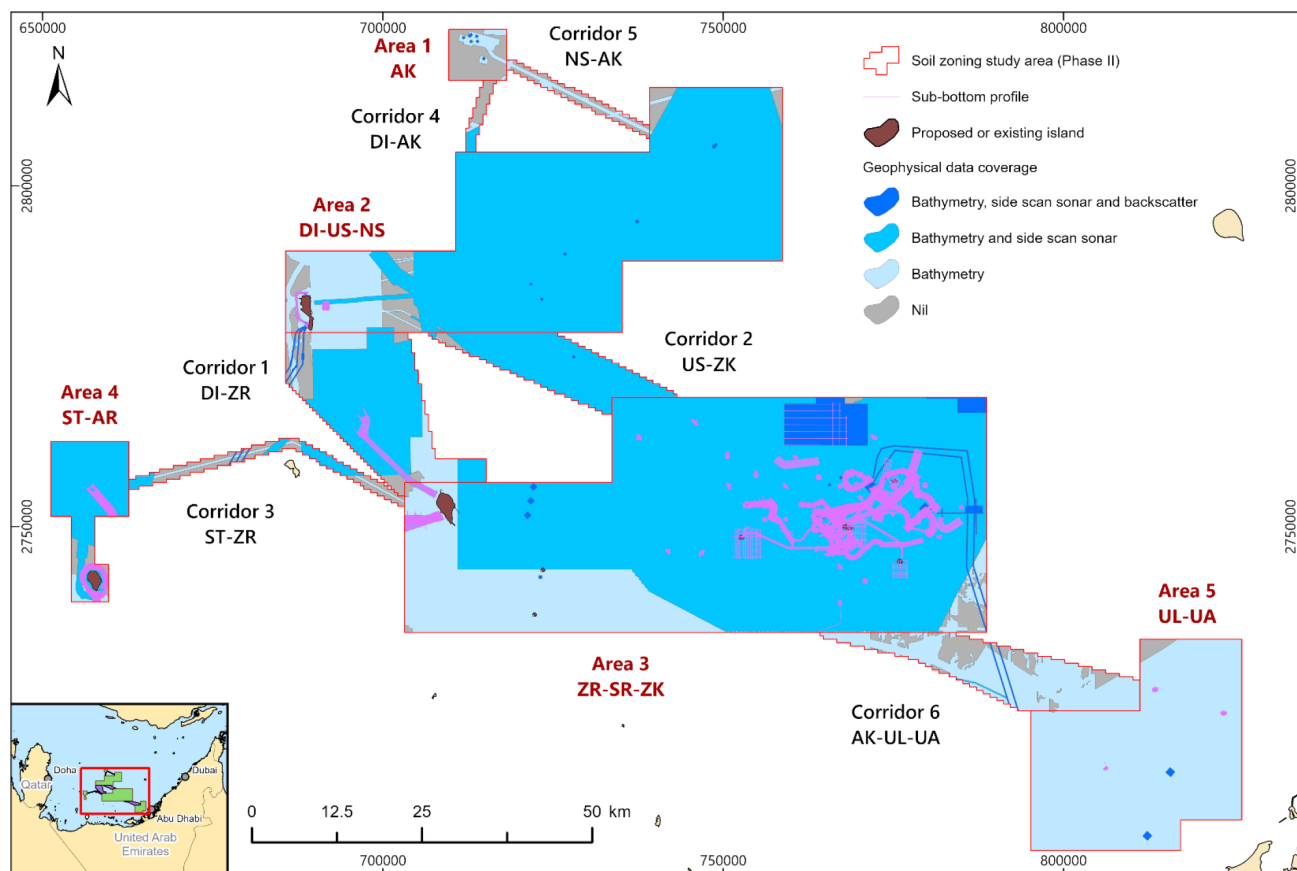


Figure 3—Geophysical data coverage

Geotechnical Data

A large quantity of offshore geotechnical data was compiled for the characterization of seabed within the study area and for the selection of geotechnical parameters for the PSI/CSI analyses conducted for this study. These geotechnical data were acquired by geotechnical surveys conducted between 1970s and 2023, encompassing data of varying qualities. The collected geotechnical data includes, but is not limited to, borehole or sampling locations and coordinates, lithology and soil descriptions, laboratory test results and cone penetration test (CPT) results. Among the available offshore geotechnical data, data obtained from 815 offshore CPTs (including piezocone penetration tests and thermal CPTs) and 3269 offshore sampling locations (including boreholes, vibrocores, drop cores etc.) that are relevant for PSI/CSI study have been selected for inclusion to the database for this study. The spatial distribution of the selected CPTs and offshore sampling locations is shown on Fig. 4.

At offshore investigation sites where soil samples were collected and sampling techniques were recorded, the sampling methods used to collect both uncemented and cemented sediments within the upper 1 meter of the seabed were examined. Since grab samplers generally only recovers superficial and heavily disturbed sediment samples that are not representative of the sediment's in-situ conditions, seabed information gathered from the grab samples were ignored for seabed characterization in this study. Deeper sediment samples collected using vibrocorer, hammer sampler, split spoon sampler, and cable percussion methods are also typically disturbed. Therefore, the geotechnical information gathered from these samples was mainly used for sediment type classification purposes in this study. In contrast, the uncemented seabed sediment samples obtained through piston sampling, push sampling, and drop coring are generally less disturbed and more representative of the sediment's in-situ conditions. As such, laboratory test data for uncemented sediment samples recovered by these sampling methods were prioritized for the selection of geotechnical parameters for PSI/CSI analysis.

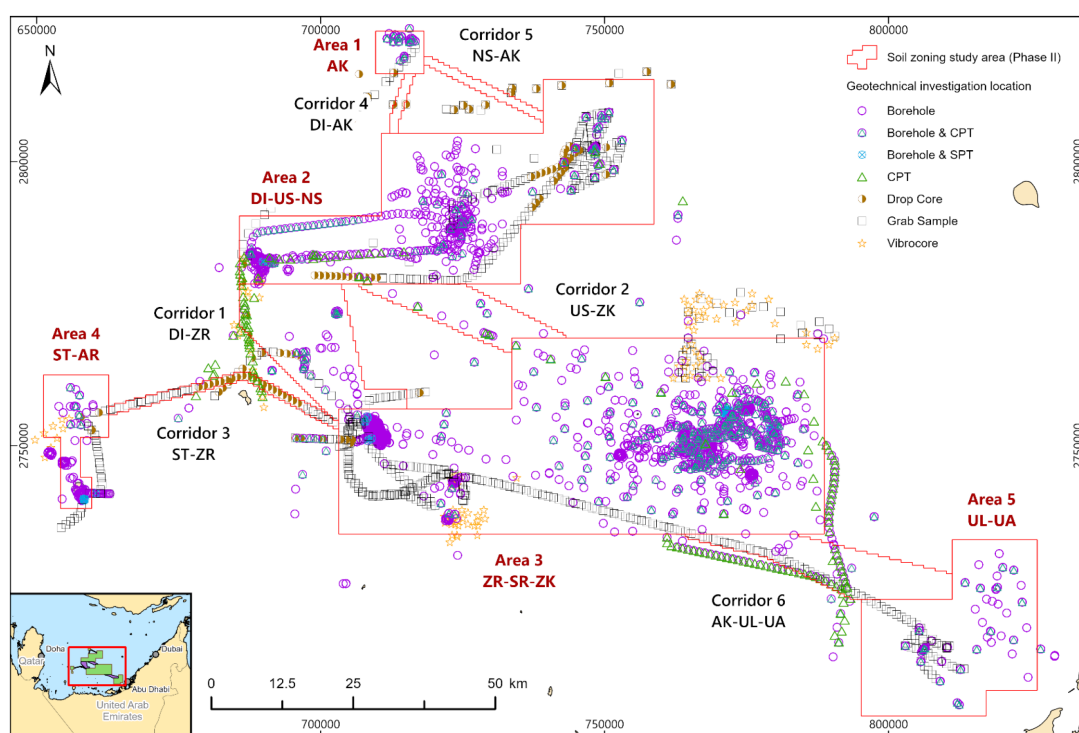


Figure 4—Geotechnical investigation locations included in the database

The borehole and in-situ test data collated for the database include (i) sediment descriptions provided in borehole/sample logs; (ii) Total Core Recovery (TCR), Solid Core Recovery (SCR) and Rock Quality Designation (RQD) values for rock cores; (iii) CPT data; and (v) Standard Penetration Test (SPT) data. The laboratory test data collated for the database include (i) basic soil classification data (e.g. water content, dry density, specific gravity, bulk density, Atterberg limits, and particle size distribution); (ii) rock strength measurements; (iii) carbonate content; (iv) index undrained shear strength test data; (v) soil friction angles; and (vi) rock/soil-interface shear resistance measurements.

Subsea Pipeline and Cable Parameters

To generate a monotonic PSI/CSI parameter database for the development of PSI/CSI calculators to estimate monotonic PSI/CSI parameters appropriate for pipeline and cable design, it is essential to gather a comprehensive range of realistic pipeline and cable parameters for the Monte Carlo PSI/CSI analyses conducted for this study. These parameters include total outer diameter, bending stiffness, submerged weights under empty, flooded, and operational conditions for pipelines, submerged weight for cables, outer surface roughness, as well as the typical installation parameters such as lay method, lay rate, and bottom lay tension.

In this study, data for typical pipeline and cable parameters were extracted from the pipeline and cable design reports for previous ADNOC projects. During the data collection and compilation process, the data was carefully assessed, and systematically screened and compiled for the establishment of realistic pipeline/cable parameter ranges for the Monte Carlo PSI analyses for rigid pipelines with nominal diameter ranges from 4-inches to 48-inches; flexible pipelines with nominal diameter ranges from 4-inches to 16-inches and cables with outer diameter ranges from 25 mm to 275 mm.

The Monte Carlo PSI/CSI analyses conducted to generate the PSI/CSI parameter database assumed that the rigid pipelines are coated with fully rough standard impinged concrete weight coating while the flexible pipelines are coated with smooth HDPE/PP coating and the outer serving for the cables can be either smooth HDPE or PP yarn. The rigid pipelines are assumed to be S-laid while the flexible pipelines and cables are assumed to be reel-laid.

Data Digitization

The majority of geotechnical data retrieved from the ADNOC archive are in the form of report format. Earlier reports exist as scanned PDF files created from physical documents, while more recent reports were available as native electronic PDF files. Therefore, the subsequent step in the data compilation process involved digitizing these reports to extract geotechnical data relevant for this study.

Optical character recognition (OCR) software was used to digitize geotechnical data from these PDF reports into a digital geotechnical database format. When compared to manual digitization, this process improved data digitization efficiency and reduced the likelihood of transcription errors. In addition, the native geotechnical database provided by survey contractors for several site investigations within the study area were also converted into a consistent geotechnical database format to facilitate data integration.

The digitized data was methodically integrated into geotechnical database spreadsheets, forming an extensive digital geotechnical database for the study area. When hosted within a GIS platform, the digital database enables streamlined access to geotechnical information at investigation sites across the study area and features customizable data viewing capabilities to accommodate the unique needs of different database users. Charts, tables and images were also converted into digital form (where possible, as geophysical data can be found difficult to digitize from PDF report) and georeferenced within the GIS database for use in the data integration process. Fig. 5 shows an example of viewing the digitized geotechnical information for an investigation site in ArcGIS platform.

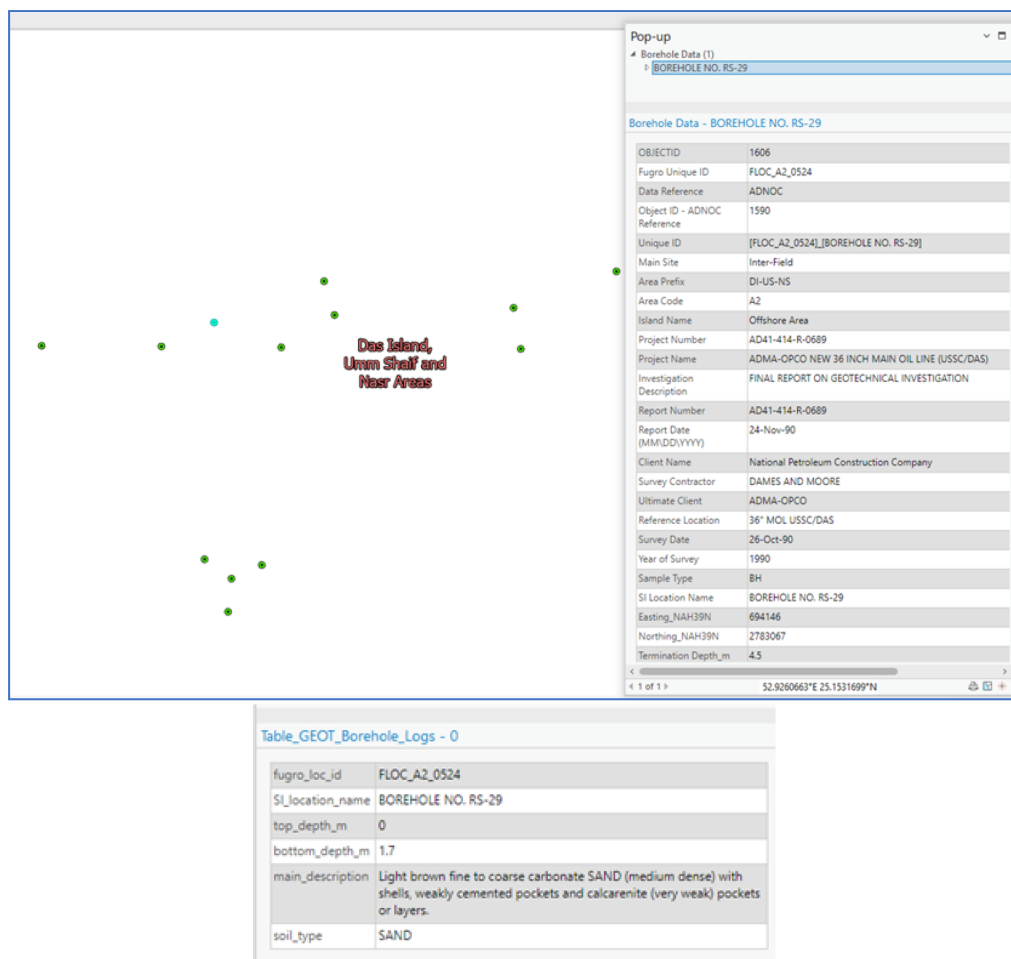


Figure 5—Example of viewing the digitized geotechnical information for an investigation site in ArcGIS platform

Establishment of Seabed Zonation and Geotechnical Parameters for PSI/CSI Analysis

Data Scoring for PSI/CSI Study

A data scoring system was established to identify good quality data that are relevant for PSI/CSI study. This data scoring system allows the quantification of uncertainty and confidence levels in the interpreted seabed conditions and the assessed PSI/CSI parameters, as well as a better understanding of data gaps in the existing database. Within this system, data with high score were given greater emphasis and weighting during integration and interpretation for seabed zonation and characterization. Conversely, data identified as anomalous were excluded from the analysis to ensure robustness and reliability of the study outcomes.

Geophysical Data Scoring. In this study, the quality of the gridded geophysical data (i.e., MBES bathymetry, backscatter and SSS) and sub-bottom profiles compiled in the database were assessed and scored using a data scoring matrix developed based on data resolution, data coverage and the presence of artefacts. Fig. 6 illustrates the importance of scoring the geophysical data for seabed interpretation by comparing three bathymetry datasets from the same area but with different data quality. The left panel displays data exhibiting TIN artefacts (polygonal surfaces); the middle panel presents data with a 1 m grid size and 300 m gaps between swaths; and the right panel illustrates continuous data with a 2 m grid size. As shown on Fig. 6, the resolution and artifacts of gridded bathymetry data could affect seabed interpretation because features smaller than the grid size may not be resolved and could lead to uncertainty in the identified seabed conditions relevant to PSI/CSI study.

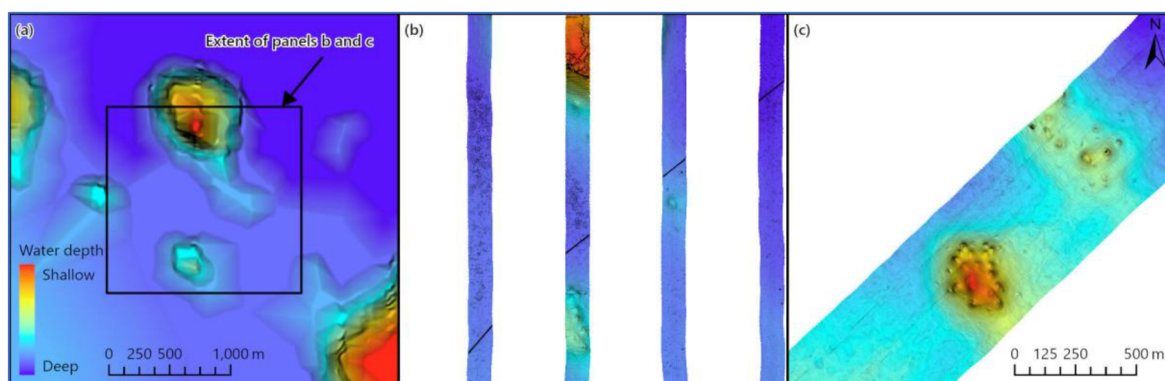


Figure 6—Comparison of co-located bathymetry data in Area 2. (a) water depth with 10 m grid size and TIN artifacts; (b) Swaths of water depth 100 m wide and spaced 300 m apart with 1 m grid size; (c) Water depth with 2 m grid size

Geotechnical Data Scoring. In this study, the quality of geotechnical data for each investigation site were assessed and scored using a data scoring matrix developed based on (i) the certainty of location information; (ii) report status and sample logging quality; (iii) sampling method used; (iv) the availability and quality of in-situ and laboratory tests within the top 1 m of seabed that are relevant for PSI/CSI study; and (v) the recency of data. These data scoring criteria collectively quantify the confidence and uncertainty levels in the interpreted seabed conditions and the geotechnical parameters selected for PSI/CSI analysis.

Relative Confidence Level Map for PSI/CSI Study. Based on the geophysical and geotechnical data quality score, a relative confidence level map was developed. The relative confidence levels are related to the aggregated data quality score of the available geotechnical and geophysical data at each investigation location or survey area. At location or area where multiple datasets are present, the data with the highest score was used for seabed characterization. A distinction of the relative confidence level is made for the geotechnical and geophysical data because different scoring criteria were used for each data type.

Based on the aggregated data quality score and the PSI/CSI specific seabed characterization requirements, the geophysical data were classified into 15 relative confidence levels while the geotechnical data were classified into 4 relative confidence levels, as shown on Fig. 7. A higher relative confidence level number for both geophysical and geotechnical data corresponds to higher confidence level in the interpreted seabed zones based on relatively higher data quality while lower relative confidence level number suggest reduced confidence.

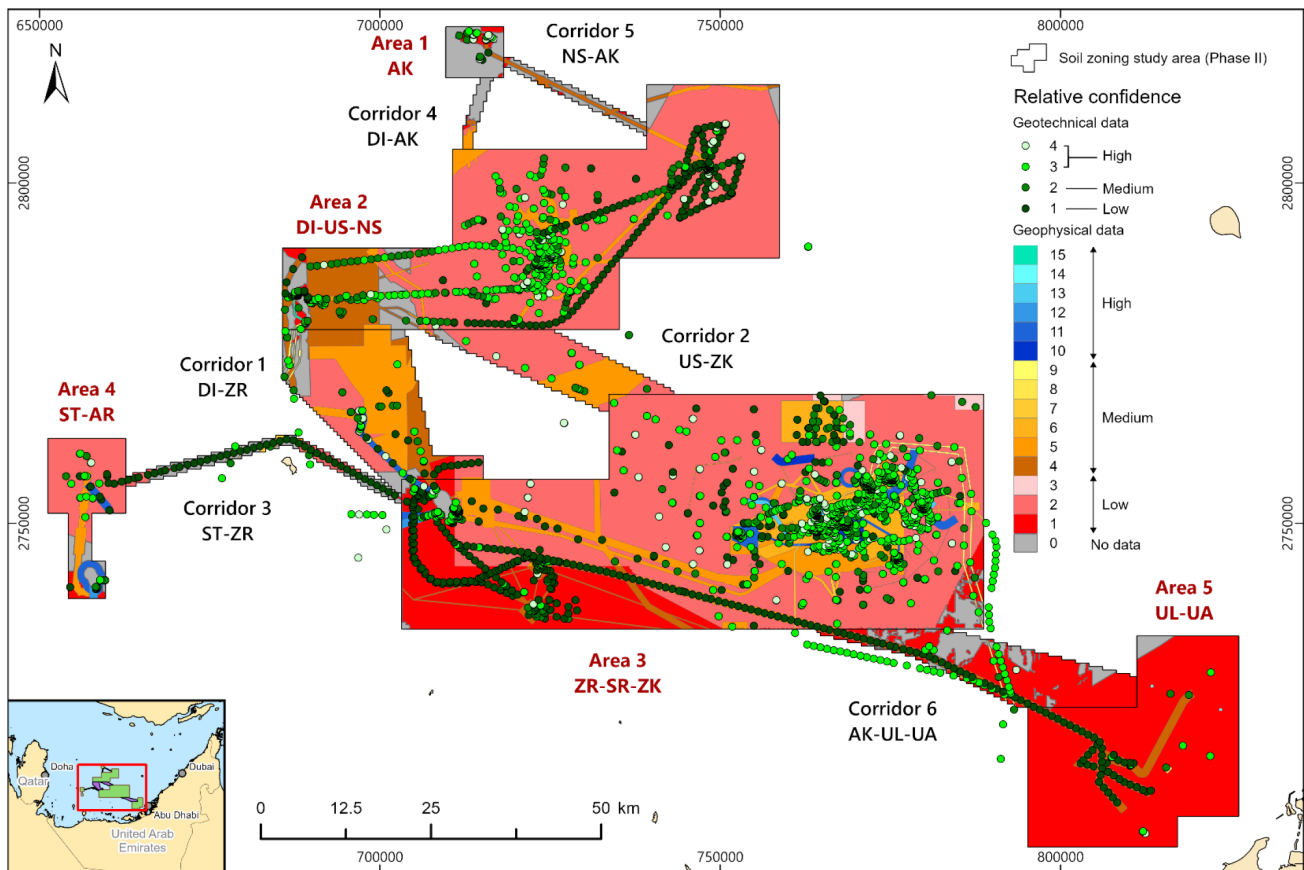


Figure 7—Relative confidence level map of the compiled geophysical and geotechnical dataset for the PSI/CSI study purposes

The defined relative confidence level for geophysical and geotechnical data were incorporated into the study area boundaries to produce a map that presents the relative confidence level of the seabed interpretation across the study area and the PSI/CSI parameters generated by the PSI/CSI calculator developed in this study, as shown on Fig. 7. This map could be used for planning new geophysical and geotechnical surveys for new pipeline and cable routes to increase confidence level in PSI/CSI seabed zonation and parameters. In areas where existing data are of low to medium relative confidence levels, comprehensive route-specific PSI/CSI targeted data acquisition should be performed along the proposed pipeline and cable routes to fill in the data gaps and reduce design uncertainty. In areas with high relative confidence geophysical and geotechnical datasets, a targeted reconnaissance geophysical survey is recommended to confirm the seabed has not changed either naturally or anthropogenically since the latest survey was conducted; and a targeted geotechnical survey is recommended to recover seafloor samples for PSI/CSI specific laboratory testing.

Data Integration and Interpretation Methodology for PSI/CSI Specific Seabed Zonation

The evaluation of seabed and shallow subsurface conditions to define PSI/CSI specific seabed zonation for this study involves a thorough assessment of all data integrated into the study's database. This integrated data

analysis approach offers greater reliability and accuracy than conventional seabed classification methods that rely on a single database (e.g., side scan sonar, SSS) or a limited subset of geophysical or geotechnical data. Reliable outcomes through this approach are achieved through the rigorous interpretation of all available data by geophysical and geotechnical specialists working in close collaboration. Fig. 8 shows a schematic diagram for the data integration and interpretation methodology that adopted in this study.

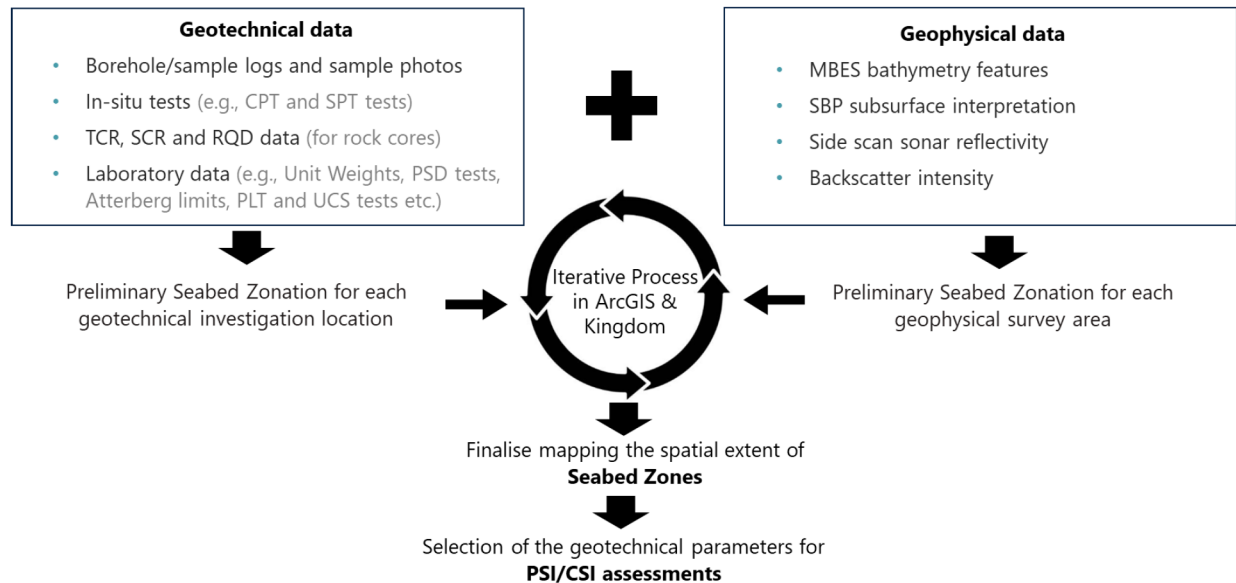


Figure 8—Data integration and interpretation methodology

As shown on Fig. 8, the interpretation of the integrated geophysical and geotechnical datasets is an iterative process that entails thorough review and analysis of geotechnical, geological, and geophysical information. This process typically involved multiple consultations and discussions between geophysical and geotechnical specialists to ensure the interpretations of the integrated geotechnical and geophysical dataset are consistent, rigor and accurately reflect the actual seabed conditions.

In this study, geotechnical data within the top 1 m of the seabed such as the CPT/PCPT results, SPT results, sampling method, sediment descriptions, sample photos, TCR/SCR/RQD values for rock, and laboratory test results were reviewed and analyzed to identify the dominant surficial sediment type and its thickness at each geotechnical investigation location. Subsequently, each location was assigned a preliminary seabed zone ID and the data were imported into ArcGIS and Kingdom platforms to compare with the preliminary seabed conditions interpreted from the geophysical dataset. To delineate the spatial extent of seabed zones with similar geotechnical characteristics, multiple rounds of cross-validation were conducted between geotechnical and geophysical interpretations, ensuring coherent seabed condition interpretation for PSI/CSI study purposes.

Due to the data quality variability, the approach used to interpret the integrated geophysical and geotechnical dataset has been adapted based on the data quality and availability across the study area. In areas with extensive high-quality datasets and previous PSI/CSI studies, the data interpretation and seabed zonation process started by reviewing past reports and reassessing historical seabed zones using the integrated geophysical and geotechnical dataset. The previous seabed feature mapping and zonation were updated based on the additional dataset compiled in this study.

In areas with medium geophysical data score (see Fig. 7) where sub-bottom profiler (SBP) is typically not available and subsurface layering cannot be determined, the seabed zonation for PSI/CSI study was primarily defined based on the interpretation of seabed geophysical data (i.e., MBES and SSS), as well as seafloor and subsurface geotechnical data (box cores, boreholes, PCPTs, and laboratory testing), where

available. In areas with limited bathymetry data, seabed zonation was guided by the observed seabed features within the available bathymetry data, supplemented by geotechnical data where possible. This approach inevitably results in interpretation with the lowest confidence level. This applies to areas with either low bathymetry data score or a combination of low bathymetry data, SSS and/or backscatter score (see Fig. 7).

An example is shown on Fig. 9 to illustrate the data integration and interpretation process adopted for an area with high geophysical and geotechnical data score to demonstrate the superiority and the reliability of the integrated data analysis over conventional approach that based on a single dataset. In this area, paired borehole and PCPT data were acquired at discrete locations while geophysical data (e.g., MBES bathymetry, side scan sonar, and SBP data) were acquired as continuous measurements across the survey area. Both geotechnical and geophysical data were integrated, reviewed, and compared to interpret seabed and shallow subsurface conditions between the geotechnical investigation locations within this area.

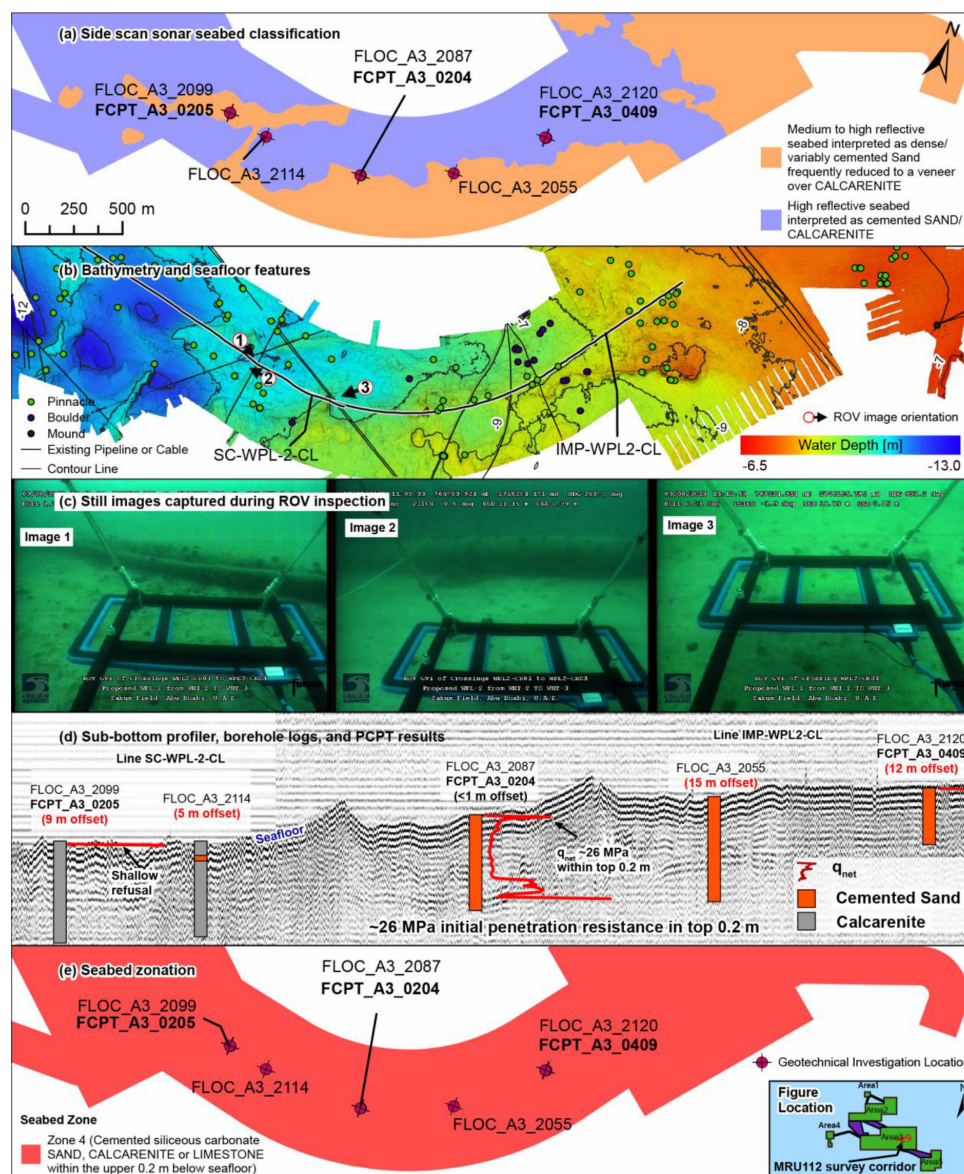


Figure 9—Maps illustrating (a) seabed classification interpreted from side scan sonar data alone; (b) seabed bathymetry; (c) still images captured during ROV inspection; (d) sub-bottom profiler with PCPTs, boreholes, and subsurface interpretation, and; (e) seabed zone for PSI/CSI assessment purposes defined in this study

As shown in Fig. 9a, the seafloor within the survey corridor was interpreted as ‘dense or variably cemented sand’ and ‘cemented sand or calcarenite’ based on the interpretation of side scan sonar reflectivity only (without subsurface information) within the survey corridor. In addition, the SBP shown in Fig. 9d yielded a strong seafloor reflector with weak and intermittent subsurface reflectors, which may be interpreted to represent potential uncemented material over variably cemented material (i.e., density and strength variability). However, the three PCPTs conducted at locations FCPT_A3_0205, FCPT_A3_0204 and FCPT_A3_0409 (as shown on Fig. 9d) measured seafloor refusal or maximum net cone penetration resistance (q_{net}) of over 20 MPa in the upper 0.2 m of the seafloor, suggesting cemented material at the seafloor. In addition, the five boreholes, conducted at locations FLOC_A3_2099, FLOC_A3_2114, FLOC_A3_2087, FLOC_A3_2055 and FLOC_A3_2120 also recorded cemented carbonate sediments or calcarenite in the top 1 m of seabed. Therefore, coupled with the rugged seafloor features observed in the bathymetry data in Fig. 9b, the integrated geophysical and geotechnical dataset suggests the seafloor within the survey corridor is interpreted to only consists of cemented carbonate sediment or calcarenite, instead of a mix of uncemented and cemented seafloors interpreted based on SSS or SBP interpretation only. This interpretation is further confirmed by the still images acquired by the ROV survey conducted at pipeline crossing locations (see Fig. 9c) that show the existing pipeline and cable were sitting on the seafloor with negligible embedment and no apparent sign of significant uncemented sediments was observed on the seabed floor. Therefore, the seabed within this survey corridor was reinterpreted as a single cemented sediment zone for PSI/CSI assessment purposes.

Seabed Zonation for PSI/CSI Assessment

Following the interpretation of the integrated geophysical and geotechnical dataset using the methodology described in previous section, four distinct seabed zones for PSI/CSI assessment purposes were identified. Table 2 summarizes the general descriptions of the surficial sediments in each identified seabed zone and the spatial distribution of each identified seabed zones is presented in Fig. 10. Note that the seabed zones presented in Fig. 10 were defined based on the geophysical and geotechnical characteristics of the surficial sediment unit within the top 1 m of the seabed or above the first reflector R1, whichever is shallower. Therefore, they did not account for the complete vertical variability of sediment properties with depth or any underlying geological or geotechnical units. Therefore, the defined seabed zones are not suitable for engineering applications other than for the PSI/CSI assessment for surface laid pipelines and cables.

Table 2—General sediment descriptions for each seabed zone identified within the study area

Seabed Zone ID	Description
1	Siliceous carbonate and carbonate CLAY/SILT to Sandy CLAY/SILT
2	Siliceous carbonate and carbonate Silty/Clayey SAND
3	Siliceous carbonate and carbonate slightly Silty/Clayey SAND to siliceous carbonate and carbonate clean SAND
4	Cemented siliceous carbonate SAND, CALCARENITE or LIMESTONE within the upper 0.2 m below seabed

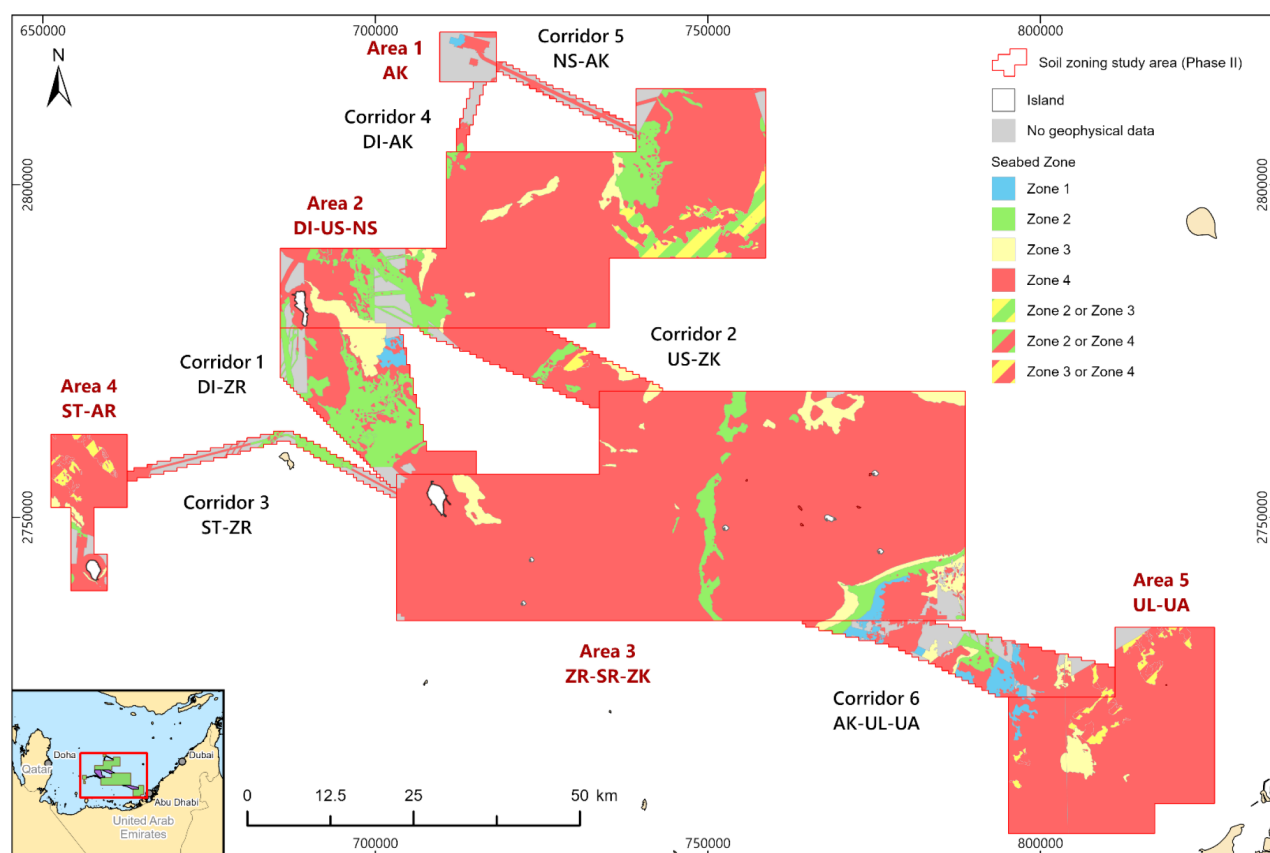


Figure 10—Seabed zones for PSI/CSI assessment purposes

The boundaries between the identified seabed zones were delineated to best group seabed with similar geotechnical conditions and characteristics that are relevant for the assessment of PSI/CSI parameters. The transitions between the seabed zones (as presented in Fig. 10) can be distinct where abrupt changes are supported by continuous and high-resolution data and can be gradational where no distinct geophysical boundary could be identified and/or a geotechnical property change could be observed between sampling locations. In some areas, the available geotechnical data is insufficient to confidently define a single seabed zone. In these areas, the data uncertainty is high enough that more than one seabed zone could be defined. This is particularly true for areas where geophysical data are available, but with no geotechnical data for reliable seabed characteristics identification. In these areas, two possible seabed zones were assigned to capture the uncertainty (i.e., "Zone 2 or Zone 3", "Zone 2 or Zone 4", and "Zone 3 or Zone 4").

The seabed zone map presented in Fig. 10 should be used in conjunction with the relative confidence level map presented in Fig. 7 to assess the confidence level of each defined seabed zone and to determine if additional investigation work is required to reduce the uncertainty of the defined seabed zone for the design of pipelines and cables within the study area.

Geotechnical Parameters for PSI/CSI Analysis

The geotechnical data for each identified seabed zone was grouped and subjected to statistical analysis and interpretation to establish geotechnical parameter bounds for the Monte Carlo PSI/CSI analyses for surface laid pipelines and cables. In instances where PSI/CSI analysis specific geotechnical parameter was absent from the compiled database, the input parameter bounds were determined based on specialist consultant's regional experience for similar soil type. Consequently, the ranges of the selected geotechnical input parameters were suitably wide to cover the uncertainty in the selected parameters (due to data sparsity)

and the potential seabed variability across the field. The type of geotechnical parameters selected for the PSI/CSI analyses conducted for this study are summarized in Table 2.

Since the uplift and axial resistances for buried pipelines are dependent on the properties of backfill materials (either natural or engineered backfills), the geotechnical parameters for their calculations are not selected based on the geotechnical data in the database. The PSI calculator for buried pipelines calculates the uplift and axial resistances based on user defined geotechnical parameters for the backfill materials selected for the project.

Table 3—Geotechnical parameters for PSI/CSI analysis

Geotechnical Parameter	Rationale
Effective unit weight, γ'	Effective unit weight will be used to capture the effects of buoyancy during pipeline/cable embedment and to calculate the seabed resistance during lateral pipeline/cable breakout
Void ratio, e	Void ratio is a soil input for the estimation of soil small strain stiffness for the calculation of pipeline dynamic seabed stiffness.
Net cone resistance, q_{net}	To estimate the undrained shear strength (s_u) profiles, which will be required for the calculations of: • Pipeline/cable embedment. • Lateral seabed resistance • Axial seabed resistance
Undrained shear strength, s_u	To derive design soil undrained shear strength profiles (ideally through calibration with penetration resistance profile measured by in-situ penetration test), which will be required for the calculations of: • Pipeline/cable embedment. • Lateral seabed resistance • Axial seabed resistance
Strength sensitivity, S_t or remoulded undrained shear strength, s_{u-rem}	To derive design remoulded undrained shear strength profiles, which will be required for the calculations of: • Pipeline/cable embedment
Coefficient of consolidation, c_v	• To allow drainage assessment during pipe lay and pipeline lateral breakout. • To support the calculations of axial seabed resistance.
Friction angle, ϕ' or interface friction coefficient, μ at low vertical stress	To derive drained interface friction coefficient design range, which will be required to support calculations of: • Lateral seabed resistance • Axial seabed resistance
Undrained interface strength ratio, $s_{u,int, res}/\sigma'_{h0}$ at low vertical stress	To derive undrained interface friction characteristics, which will be required to support calculations of: • Axial seabed resistance
Rock interface friction coefficient, μ	To derive drained rock-pipe/cable interface friction coefficient design range, which will be required to support calculations of: • Lateral seabed resistance • Axial seabed resistance

Development and Functionality of PSI/CSI Calculators

This section presents the PSI/CSI calculators developed for generating monotonic PSI/CSI parameters for the preliminary design of pipelines and cables within the study area. These calculators utilize Microsoft Excel spreadsheets as their intuitive user interface and are designed to be standalone tools executable in Microsoft Excel without the need for additional third-party software. The calculators are user-friendly and capable of generating realistic PSI/CSI parameters in seconds. Therefore, these calculators are efficient tools that can be used by pipeline and cable engineers to gain early clarity on the most effective pipeline and cable design strategies that save cost and improve project outcomes.

In this study, more than 20,000 Monte Carlo PSI/CSI analyses (each with at least 10,000 calculations) were performed using methodology draws upon findings from a wide body of research on PSI in published

technical papers, from the SAFEBUCK JIP (the findings from which have been incorporated in DNV-RP-F114, 2021), and from specialist consultant's extensive experience and work in PSI for carbonate sediments (e.g., Bransby et al., 2015; Abdel-Hakim et al., 2018; Bransby et al., 2020). The Monte Carlo analyses were conducted using the pipeline/cable inputs selected from the compiled pipeline/cable property database and the geotechnical parameters selected for each identified seabed zone. The analyses assume no sediment mobility occurs after pipe lay, prior to pipeline displacement due to instability caused by hydrodynamic forces and thermal/pressure induced buckling.

With the results generated from the Monte Carlo PSI/CSI analyses, an extensive database of monotonic PSI/CSI parameters was established. This database forms the backbone of the PSI/CSI calculators to generate low estimate (LE), best estimate (BE) and high estimate (HE) values of the monotonic PSI/CSI parameters that are appropriate for preliminary design of pipelines and cables within the study area. The relative confidence level of the calculated PSI/CSI parameters can be evaluated using the relative confidence level map presented in Fig. 7 to determine if additional investigation work is required to reduce the uncertainty of the calculated PSI/CSI parameters for pipeline and cable design.

The PSI/CSI calculators aim to cover the PSI/CSI analysis for a wide range of pipeline and cable options. Due to the nature of this study, it is impossible to determine accurate pipeline/cable installation and operational conditions (which are project dependent) for the PSI/CSI analysis for each pipeline and cable option. To account for the installation and operational condition uncertainties and their effects on the soil drainage conditions during PSI/CSI events (e.g., pipelay and first breakout due to lateral buckling), the following drainage conditions were considered in the calculation of as-laid embedment and friction factors for pipelines and cables in each identified seabed zone:

- Both undrained and drained drainage conditions were considered in the PSI/CSI analyses for surface-laid pipelines/cables in Seabed Zone 1 and Zone 2. Therefore, both undrained and drained PSI/CSI parameters are generated by the PSI/CSI calculators for surface-laid pipelines and cables in these seabed zones.
- Only drained drainage conditions were considered in the PSI/CSI analyses for surface-laid pipelines/cables in Seabed Zone 3 and Zone 4 due to the high permeability nature of the sediments in these seabed zones. Therefore, only drained PSI parameters are generated by the PSI/CSI calculators for surface-laid pipelines and cables in these seabed zones.
- The buried pipeline PSI calculator is developed to calculate both undrained and drained uplift and axial resistances for pipelines buried in natural and engineered backfill materials. The user to choose the appropriate calculation method based on the backfill material properties.

Surface-laid Pipeline and Cable PSI/CSI Calculators

The surface-laid pipeline and cable PSI/CSI calculators are designed to generate monotonic PSI/CSI parameters for surface-laid pipelines and cables based on user defined pipeline/cable inputs. The calculators comprise of three main components listed below and shown in Fig. 11:

- User interface in Microsoft Excel spreadsheet
- Calculation module as a Dynamic Link Library
- SQLite databases of precomputed monotonic PSI/CSI parameters.

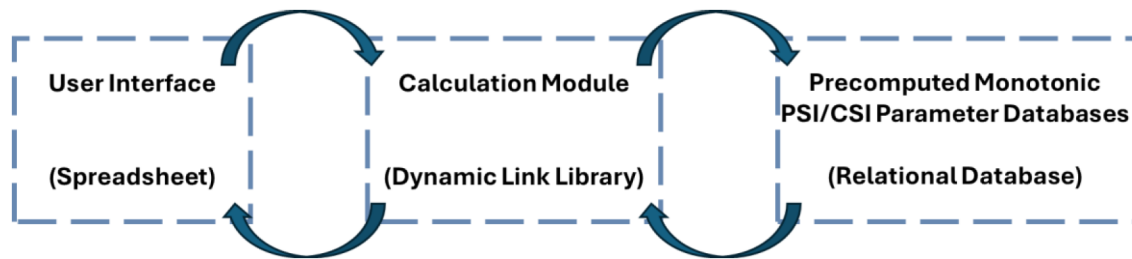


Figure 11—Components of surface-laid pipeline and cable PSI/CSI calculators

The user interface Excel spreadsheet allows user to input pipeline/cable inputs. When a calculation is initiated, the VBA code within the user interface spreadsheet will initiate the PSI/CSI calculation functions built in the calculation module DLL, which was created using C++ programming language. The calculation module will search through the precomputed monotonic PSI/CSI parameter databases for a set of relevant precomputed PSI/CSI parameter dataset based on the user defined pipeline/cable inputs. The calculation module will then perform statistical analyses on the selected PSI/CSI parameter dataset and calculate the LE, BE and HE monotonic PSI/CSI parameters based on user defined pipeline/cable inputs using multi-dimensional linear interpolation method. The calculated LE, BE and HE monotonic PSI/CSI parameters will finally be outputted and presented in the user interface Excel spreadsheet.

The precomputed monotonic PSI/CSI parameter databases were created using the Monte Carlo PSI/CSI analysis results following the steps described below:

1. Use the compiled property database for rigid pipelines, flexible pipelines and cables to define a pipeline/cable input combination matrix for Monte Carlo PSI/CSI analyses.
2. Perform Monte Carlo PSI/CSI analyses using the pipeline/cable input combinations defined in the combination matrix and the geotechnical parameters selected for each seabed zone identified within the study area. Each Monte Carlo PSI/CSI analysis involved at least 10,000 calculations. With the Monte Carlo PSI/CSI analysis results, monotonic PSI/CSI parameter databases were created.
3. Validate the calculator outputs by comparing the calculator outputs with the Monte Carlo PSI/CSI analysis outputs generated using specialist consultant in-house PSI/CSI calculation tool.
4. Refine the pipeline/cable input combination input matrix (such as add more analysis cases) based on the outcome of Step 3 and repeat Step 1 to Step 3 interactively until the calculators are able to generate PSI/CSI parameters with reasonable accuracy when compared to the Monte Carlo PSI/CSI analysis outputs generated by specialist consultant's in-house PSI/CSI calculation tool.

The PSI calculators have a built-in feature that checks the user's pipeline/cable inputs against the pipeline/cable property database used to generate the monotonic PSI/CSI parameter database. Should the user's pipeline inputs fall outside the pipeline/cable property database, messages will be shown in the user interface Excel spreadsheet to guide the user on the appropriate pipeline/cable inputs. This feature ensure the calculated monotonic PSI/CSI parameters are appropriate for the intended pipeline/cable analysis. This feature also prevents user from inputting unrealistic pipeline/cable inputs, which could result in unrealistic PSI/CSI parameters being calculated.

In this study, seven calculators were developed for each surface-laid asset type - surface-laid rigid pipeline, flexible pipeline and cable. This comprises of four calculators for Seabed Zone 1 to Zone 4 (one for each seabed zone) and three calculators for areas identified with two possible seabed zones (i.e., Seabed Zone 2 or Zone 3, Seabed Zone 2 or Zone 4, Seabed Zone 3 or Zone 4). An example of the PSI calculator inputs and outputs for a surface laid rigid pipeline is shown in [Fig. 12](#).

Pipe Parameter			
Parameter	Symbol	Unit	Value
Nominal Diameter	ND	inch	26
Total Outer Diameter (Including All External Coatings)	OD	m	0.97
Bending Stiffness	EI	kN*m ²	463201
Submerged Empty Weight	W' _{empty}	kN/m	6.36
Submerged Flooded Weight	W' _{flooded}	kN/m	9.47
Submerged Dewatered Weight	W' _{dewatered}	kN/m	6.36
Submerged Operational Weight	W' _{op}	kN/m	6.55
Minimum Bottom Lay Tension	T _{0,min}	kN	750
Maximum Bottom Lay Tension	T _{0,max}	kN	1000
Ratio of Span Length to Span Support Length	L _{span} /L _{sh}	-	
Span Length	L _{span}	m	20

Normalised 'As-laid' Embedment and Pipe-lay Vertical Seabed Stiffness

Condition	Parameter	Symbol	Unit	LE	BE	HE
Empty Weight	Normalised 'As-laid' Embedment	w/D	-	0.04	0.18	0.45
	Pipe-lay Vertical Seabed Stiffness	k _{lay-down}	kN/m/m	14.5	36.0	165.3
Flooded Weight	Normalised 'As-laid' Embedment	w/D	-	0.04	0.18	0.45
	Pipe-lay Vertical Seabed Stiffness	k _{lay-down}	kN/m/m	21.6	53.4	244.2
Operational Weight	Normalised 'As-laid' Embedment	w/D	-	0.04	0.18	0.45
	Pipe-lay Vertical Seabed Stiffness	k _{lay-down}	kN/m/m	14.9	36.9	168.8

Monotonic Undrained Friction Factor

Condition	Parameter	Symbol	Unit	LE	BE	HE
Empty Weight	Undrained Lateral Breakout Friction Factor	H _{brk,u} /W' _{empty}	-	0.57	0.98	1.52
	Undrained Lateral Residual Friction Factor	H _{res,u} /W' _{empty}	-	0.36	0.67	1.27
	Undrained Axial Residual Friction Factor	T _{res,u} /W' _{empty}	-	0.25	0.42	0.60
Flooded Weight	Undrained Lateral Breakout Friction Factor	H _{brk,u} /W' _{flooded}	-	0.45	0.81	1.23
	Undrained Lateral Residual Friction Factor	H _{res,u} /W' _{flooded}	-	0.36	0.67	1.26
	Undrained Axial Residual Friction Factor	T _{res,u} /W' _{flooded}	-	0.25	0.42	0.60
Operational Weight	Undrained Lateral Breakout Friction Factor	H _{brk,u} /W' _{op}	-	0.56	0.96	1.49
	Undrained Lateral Residual Friction Factor	H _{res,u} /W' _{op}	-	0.36	0.67	1.27
	Undrained Axial Residual Friction Factor	T _{res,u} /W' _{op}	-	0.29	0.50	0.72

Normalised Displacement to Mobilise the Undrained Friction Factor

Condition	Parameter	Symbol	Unit	LE	BE	HE
Empty Weight	Normalised Undrained Lateral Breakout Displacement	U _{brk,u} /D	-	0.03	0.15	0.42
	Normalised Undrained Lateral Residual Displacement	U _{res,u} /D	-	0.11	1.10	4.28
	Normalised Undrained Axial Residual Displacement	x _u /D	-	0.001	0.005	0.500
Flooded and Operational Weight	Normalised Undrained Lateral Breakout Displacement	U _{brk,u} /D	-	0.03	0.15	0.42
	Normalised Undrained Lateral Residual Displacement	U _{res,u} /D	-	0.11	1.11	4.21
	Normalised Undrained Axial Residual Displacement	x _u /D	-	0.001	0.005	0.500

Maximum Dynamic Seabed Stiffness

Parameter	Symbol	Unit	LE	BE	HE
Maximum Vertical Dynamic Seabed Stiffness	K _{v,max}	kN/m/m	16200	62500	459500
Maximum Lateral Dynamic Seabed Stiffness	K _{l,max}	kN/m/m	11100	49600	456600

Monotonic Drained Friction Factor

Condition	Parameter	Symbol	Unit	LE	BE	HE
Empty Weight	Drained Lateral Breakout Friction Factor	H _{brk,d} /W' _{empty}	-	0.61	1.13	2.47
	Drained Lateral Residual Friction Factor	H _{res,d} /W' _{empty}	-	0.50	0.85	1.64
	Drained Axial Residual Friction Factor	T _{res,d} /W' _{empty}	-	0.55	0.96	1.86
Flooded Weight	Drained Lateral Breakout Friction Factor	H _{brk,d} /W' _{flooded}	-	0.59	1.05	2.11
	Drained Lateral Residual Friction Factor	H _{res,d} /W' _{flooded}	-	0.50	0.85	1.65
	Drained Axial Residual Friction Factor	T _{res,d} /W' _{flooded}	-	0.55	0.96	1.88
Operational Weight	Drained Lateral Breakout Friction Factor	H _{brk,d} /W' _{op}	-	0.61	1.12	2.41
	Drained Lateral Residual Friction Factor	H _{res,d} /W' _{op}	-	0.50	0.85	1.63
	Drained Axial Residual Friction Factor	T _{res,d} /W' _{op}	-	0.55	0.96	1.86

Normalised Displacement to Mobilise the Drained Friction Factor

Condition	Parameter	Symbol	Unit	LE	BE	HE
Empty Weight	Normalised Drained Lateral Breakout Displacement	U _{brk,d} /D	-	0.03	0.15	0.42
	Normalised Drained Lateral Residual Displacement	U _{res,d} /D	-	0.09	0.80	3.01
	Normalised Drained Axial Residual Displacement	x _d /D	-	0.001	0.005	0.500
Flooded and Operational Weight	Normalised Drained Lateral Breakout Displacement	U _{brk,d} /D	-	0.03	0.15	0.42
	Normalised Drained Lateral Residual Displacement	U _{res,d} /D	-	0.09	0.81	2.96
	Normalised Drained Axial Residual Displacement	x _d /D	-	0.001	0.005	0.500

Figure 12—Example of surface laid rigid pipeline PSI calculator inputs and outputs

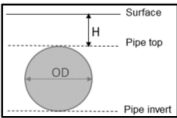
Buried Pipeline PSI Calculator

The buried pipeline PSI calculator developed in this study estimates uplift and axial resistances in accordance with DNV-RF-F114 (DNV, 2021). Since the geotechnical parameters for estimating the uplift and axial resistances are dependent on the properties of the backfill material (either natural or engineered backfill), the buried pipeline PSI calculator calculates these resistances based on user defined backfill properties. The calculator consists of the following components:

- User interface in Microsoft Excel spreadsheet
- Calculation module as a Dynamic Link Library

Similar to the PSI calculator for surface laid pipeline, the buried pipeline PSI calculator uses Microsoft Excel spreadsheet as its intuitive user interface for user to input pipeline and geotechnical parameters. When a calculation is initiated, the VBA code within the user interface spreadsheet will initiate the calculation functions built in the calculation module DLL to calculate the uplift and axial resistances based on user inputs. The calculated LE, BE and HE uplift and axial resistances will then be outputted and presented in the user interface spreadsheet. The calculator outputs were validated by comparing its results against the specialist consultant's in-house calculation tool and making sure that both generate identical results for identical pipe parameter and backfill property inputs. An example of the buried pipeline PSI calculator inputs and outputs is shown in Fig. 13.

Pipe Parameter								
Parameter	Symbol	Unit	Value					
Total Outer Diameter	OD	m	1.4					
Submerged Pipe Weight	W'	kN/m	5					
Soil Cover Height Above Pipe	H	m	2					



Backfill Material Property						
Parameter	Symbol	Unit	LE	BE	HE	
Effective Unit Weight	γ'_{fill}	kN/m ³	5	7.5	10	

Undrained Backfill Material Properties for Undrained Uplift and Axial Resistance Calculation						
Undrained Shear Strength at Seabed Surface	s_{u0}	kPa	0	1	3	
Undrained Shear Strength at Pipe Top	s_{u1}	kPa	5	8	10	
Undrained Shear Strength at Pipe Invert	s_{u2}	kPa	10	16	20	
Pipe-backfill Material Interface Adhesion Factor	α	-	0.9	1	1	

Drained Backfill Material Properties for Drained Uplift and Axial Resistance Calculation						
Friction Angle	ϕ_{fill}	degree	35	40	55	
Drained Pipe-backfill Material Interface Friction Coefficient	μ_{fill}	-	0.42	0.58	0.68	

Undrained Uplift Resistance						
Parameter	Symbol	Unit	LE	BE	HE	
Undrained Uplift Resistance	$F_{uplift,u}$	kN/m	33.8	54.6	73.6	
Undrained Uplift Friction Factor	$F_{uplift,u}/W'$	-	6.8	10.9	14.7	

Undrained Axial Resistance						
Parameter	Symbol	Unit	LE	BE	HE	
Undrained Axial Resistance	$F_{axial,u}$	kN/m	29.7	52.8	66.0	
Undrained Axial Friction Factor	$F_{axial,u}/W'$	-	5.9	10.6	13.2	

Drained Uplift Resistance						
Parameter	Symbol	Unit	LE	BE	HE	
Drained Uplift Resistance	$F_{uplift,d}$	kN/m	24.8	56.7	101.5	
Drained Uplift Friction Factor	$F_{uplift,d}/W'$	-	5.0	11.3	20.3	

Drained Axial Resistance						
Parameter	Symbol	Unit	LE	BE	HE	
Drained Axial Resistance	$F_{axial,d}$	kN/m	16.7	31.3	40.4	
Drained Axial Friction Factor	$F_{axial,d}/W'$	-	3.3	6.3	8.1	

Figure 13—Example of buried pipeline PSI calculator inputs and outputs

Integration to ADNOC ArcGIS Platform

ADNOC Offshore use ArcGIS software as a foundational platform for geospatial analysis and operational intelligence. The ArcGIS software serves as a powerful tool within the company that enable teams to visualize, analyze, and interpret spatial data in ways that drive informed decision-making. By integrating GIS into their workflows, ADNOC Offshore teams can map assets, track trends, and conduct advanced spatial analysis to optimize logistics, market strategies, and resource management. ArcGIS supports collaboration by providing interactive maps and dashboards, allowing different teams within ADNOC Offshore to share insights efficiently. Its robust capabilities empower ADNOC Offshore teams to identify opportunities, mitigate risks, and gain a competitive edge through data-driven and location-based intelligence.

This study has developed an innovative ArcGIS tool (denoted as Pipeline/Cable Alignment Segmentation Tool) designed specifically to enable extraction of information on seabed properties along any subsea pipeline or cable route within the study area. When a pipeline or cable alignment data is imported to the ArcGIS platform, the tool can automatically divide a pipeline or cable alignment into distinct segments based on the seabed zone boundaries presented in [Fig. 10](#) and provide information on the geotechnical characteristics of the seabed where each segment of subsea pipeline or cable transverses. Hyperlinks to the relevant PSI/CSI calculators will also be provided to enable pipeline and cable engineers to calculate PSI/CSI parameters for preliminary pipeline and cable design. A schematic diagram that illustrates the functionality of the Pipeline/Cable Alignment Segmentation Tool is presented on [Fig. 14](#).

The Pipeline/Cable Alignment Segmentation Tool, along with the geophysical and geotechnical database, data relative confidence level map ([Fig. 7](#)), seabed zone map ([Fig. 10](#)) and PSI/CSI calculators developed in this study were integrated into ADNOC's ArcGIS platform. This integration enables a spatially aware and streamlined pipeline and cable engineering workflow to optimize future project planning and strengthen decision-making through reliable, data-informed insights. This integration drives design efficiency, reduces manual effort, and ensures consistency in seabed zone classification and PSI/CSI assessment. By tailoring the pipeline and cable design and analysis to the specific geotechnical properties of seabed where each pipeline/cable segment transverses, the workflow enhances the robustness and safety of engineering solutions, ultimately supporting more cost-effective and resilient subsea infrastructure development.

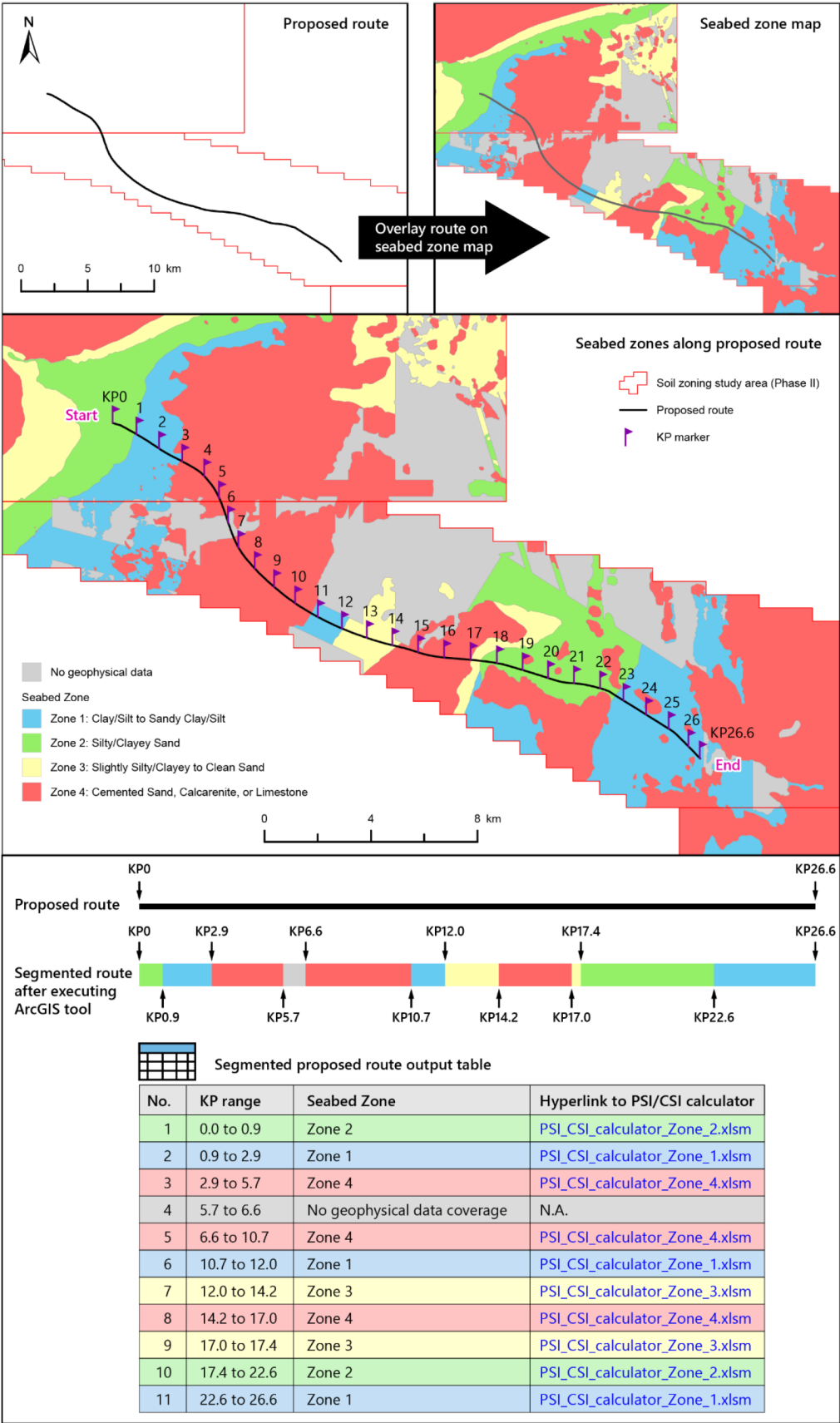


Figure 14—Schematic diagram illustrating the functionality of the Pipeline/Cable Alignment Segmentation Tool

Summary and Conclusion

An extensive geophysical and geotechnical database was created from a large quantity of geophysical and geotechnical data acquired in ADNOC's offshore fields. The database was meticulously reviewed, screened and analyzed to develop:

- A map that presents the relative confidence levels for the compiled geophysical and geotechnical datasets, the seabed zones defined across the study area and the PSI/CSI parameters generated by the PSI/CSI calculators developed in this study.
- A map that shows the spatial distribution of seabed zones with similar geotechnical characteristics for PSI/CSI study purposes
- PSI/CSI calculators for generating monotonic PSI/CSI parameters for preliminary pipeline and cable design.
- A Pipeline/Cable Alignment Segmentation ArcGIS Tool for extracting information on seabed properties along user defined subsea pipeline or cable alignment within the study area. This tool also provides hyperlinks to the relevant PSI/CSI calculators for each seabed zone that the pipeline or cable transverses.

The above maps and tools were integrated into ADNOC's ArcGIS platform to enable ADNOC's pipeline and cable engineers to enhance future project planning and make data-driven decisions in their pipeline and cable design processes. It is recommended that the databases, maps and tools are regularly updated, whenever new data and information become available.

Going forward, this integrated database can be used to:

- Optimize future geophysical and geotechnical surveys: The data relative confidence level map (Fig. 7) can be used to plan and optimize future geophysical and geotechnical surveys by limiting the survey scopes to only acquire data required to reduce uncertainties in the PSI/CSI seabed zonation and parameters for optimal pipeline and cable design.
- Optimize pipeline and cable design: The seabed zone map will provide information about the seabed conditions that the new pipeline and cable will encounter. With the preliminary PSI/CSI parameters calculated using the PSI/CSI calculators, the pipeline and cable engineers will be able to identify the governing design challenges that influence CAPEX and OPEX early in the project. This will allow the engineers to establish a targeted design strategy such as targeted surveys and engineering analyses to tackle the identified design challenges that lead to the most cost-effective design solution.
- Optimize pipeline and cable route selection. By integrating the compiled datasets with innovative pipeline/cable routing algorithm such as the stochastic modelling approach developed by [Haneberg \(2015\)](#) could allow pipeline and cable engineers to select low cost and low risk pipeline or cable routes for their projects.
- Mitigate project risk: The ArcGIS platform's robust data management and integration capabilities enable preliminary selection of pipeline/cable routes based on the compiled datasets, identified risks/hazards along the route, identified design challenges, installation constraints, seabed morphological changes over the time etc. Early awareness of these factors will allow ADNOC to proactively plan around known risks and minimizing unforeseen impacts on schedule and cost.
- Improve project schedule: The provision of realistic PSI/CSI parameters to a project needs time because several important and necessary steps need to be completed prior to geotechnical engineering calculations. These steps include data acquisition, laboratory testing and data interpretation which are necessary to ensure the calculated PSI/CSI parameters are realistic for the seabed conditions encounter along the pipeline/cable route and are realistic for the pipeline/cable

to be designed. The elimination of one (or several) of these steps using the compiled database could greatly benefit project schedule and allow PSI/CSI parameters to be provided earlier in the project.

This study also demonstrates that seabed zonation derived from the interpretation of integrated geophysical and geotechnical datasets is more robust and reliable than seabed classifications based solely on a single dataset or a subset of geophysical and geotechnical data. To ensure robust and reliable seabed zonation and geotechnical parameter selection for PSI/CSI assessment, the analysis of the integrated dataset is best performed by a multidiscipline team that include geologists, geophysicists, and geotechnical specialists.

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